NVT Monte Carlo

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Consider a system of N atoms in a volume V and at temperature T
Assume cubic volume with length $L = V^{1/3}$
The canonical partition function for this system is

$$Q = \frac{1}{N!\,\Lambda^{3N}} \underbrace{\int_0^L \cdots \int_0^L}_{3N \text{ integrals}} e^{-\beta U} \underbrace{d\vec{r}_1 \ldots d\vec{r}_N}_{3N \text{ dimensions}}$$

Probability density of a microstate is given by

$$f(\vec{r}_1, \ldots, \vec{r}_N) \propto e^{-\beta U(\vec{r}_1, \ldots, \vec{r}_N)}$$

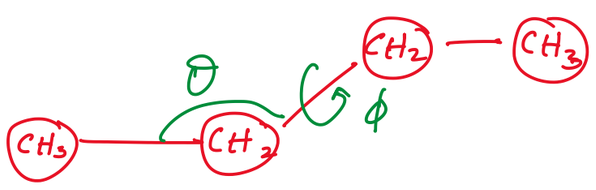
In an NVT simulation, the state is specified by the positions of all the atoms (3N real numbers between 0 and L or $-\frac{L}{2}$ and $\frac{L}{2}$)

During the simulation, one samples new configurations and then accept or reject the generated configuration according to the derived acceptance probability

In an atomic system, the most common way to generate a new configuration is to displace the atom, one at a time.

For molecular systems, new configurations are generated by (i) translation of centre of mass (ii) rotating the molecule and (iii) changing internal structure (bond length, bond angle, dihedral angle etc) of the molecule

e.g. n-Butane



<u>Condition of detailed balance</u>
Consider a change of configuration from state "o" to state "n"

$$f(o)\, \alpha(o \to n)\, acc(o \to n) = f(n)\, \alpha(n \to o)\, acc(n \to o)$$

$$acc(o \to n) = \min\left\{1, \frac{f(n)\, \alpha(n \to o)}{f(o)\, \alpha(o \to n)}\right\}$$

$$f(o) \propto e^{-\beta U_o}$$

$$f(n) \propto e^{-\beta U_n}$$

$$\therefore acc(o \to n) = \min\left\{1, e^{-\beta \Delta U} \frac{\alpha(n \to o)}{\alpha(o \to n)}\right\}$$

where $\Delta U = U_n - U_o \equiv$ change in the potential energy

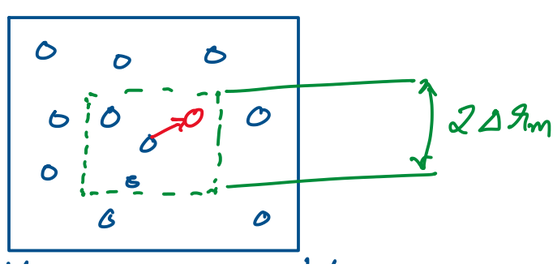
<u>Translation move</u>: Displace the position of the centre of mass of the molecule by a random amount

Let $\{x_o, y_o, z_o\}$ be the current position and
$\{x_n, y_n, z_n\}$ be the proposed position

1) Generate 3 random numbers $\{u_x, u_y, u_z\}$ uniformly between 0 and 1

2) Give random displacement to the molecule

$$\left.\begin{aligned} x_n &= x_o + 2(u_x - 1)\,\Delta r_m \\ y_n &= y_o + 2(u_y - 1)\,\Delta r_m \\ z_n &= z_o + 2(u_z - 1)\,\Delta r_m \end{aligned}\right\} \Rightarrow \alpha(o \to n) = \alpha(n \to o) = \frac{1}{8(\Delta r_m)^3}$$



$-L/2$ $L/2$

3) Apply periodic boundary condition

$$x_n = x_n - L * \text{ANINT}\left(\frac{x_n}{L}\right)$$

$$y_n = y_n - L * \text{ANINT}\,(y_n/L)$$
$$z_n = z_n - L * \text{ANINT}\,(z_n/L)$$

4) Compute ΔU

5) Accept or reject the move according to

$$\text{acc}\,(o \to n) = \min\{1,\ e^{-\beta \Delta U}\}$$

$\Rightarrow$ if $\Delta U < 0$ i.e. proposed configuration has lower energy
then $\text{acc}\,(o \to n) = 1$
i.e. move is always accepted

$\Rightarrow$ if $\Delta U > 0$, then define $q = e^{-\beta \Delta U}$

generate a uniform random number u

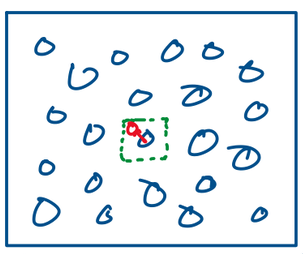
if $u < q$ then accept the proposed change

else reject the change, i.e. stay at current position

Optimum choice for Δr_m

The efficiency of the sampling phase space in an NVT simulation depends on the value of Δr_m

if Δr_m is very small



$-L/2$ $L/2$

Then the displacement would result in small changes to U
i.e. $\Delta U \approx 0$

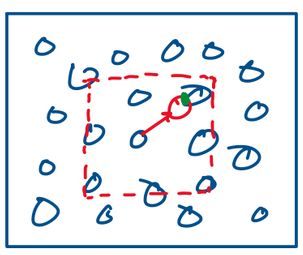
$\Rightarrow \text{acc}\,(o \to n) \approx 1$

$\Rightarrow$ most of the proposed moves will be accepted

However, the new configuration is not very different and the sampling of the phase space will be very slow.

Hence one should use a larger value of Δr_m

if Δr_m is very large



$-L/2$ $L/2$

Then it is highly likely that the proposed position will result in overlap (collision) with neighboring molecules. $\Rightarrow \Delta U \gg 0$ (a large +ve number)

Hence the acceptance of the proposed position will be very small.

This would again result in slow sampling of phase space.

Hence there exist an optimum value of Δr_m

Typically one chooses a value of Δr_m such that the probability of acceptance of your moves is around 50%.

This value of Δr_m is obtained by continuously adjusting the value of Δr_m during the initial stages of your simulation.

1) Start with a value of Δr_m

2) Perform N_T translation moves

Let N_A be the number of moves accepted

3) $\mathcal{R} = \frac{N_A}{N_T}$

if $\mathcal{R} > 0.5$ ← Δr_m is small

$\Delta r_m = 1.05\, \Delta r_m$ ← increase value of Δr_m by 5%

else ← Δr_m is large

$\Delta r_m = 0.95\, \Delta r_m$ ← decrease value of Δr_m by 5%

4) Repeat steps 2 and 3 till value of Δr_m converges

Algorithm for NVT Monte Carlo

1) Specify N, T & V (or ρ)

2) Read the initial configuration from a file

or

generate the initial configuration

3) Compute the energy U (U^{SR} and U^{LR})

4) Specify a value of Δr_m & "seed" for PRNG

5) Perform "M" Monte Carlo moves (translation move)

Typically the "M" moves are divided into cycles, where each cycle consists of N moves (one move per molecule)

∴ Number of cycles = $N_c = \frac{M}{N}$

or

$M = N_c \times N$

One either specifies a value of M (e.g. 10 million Monte Carlo moves)

or

specifies a value of N_c (e.g. 10000 Monte Carlo cycles)

5.1) Randomly choose a molecule (atom) to be displaced

$i = INT(u \times N) + 1$

5.2) Perform translation move on atom/molecule "i" ← see above for the algorithm

(5.3) Update the value of $U = \begin{cases} U_{old} + \Delta U & \leftarrow \text{if move was accepted} \\ U_{old} & \leftarrow \text{if move was rejected} \end{cases}$

(5.4) Sample properties of interest i.e. energy, pressure, excess chemical potential etc.

Store configurations at regular intervals (for post processing)

5.5) Adjust the value of Δr_m at regular intervals (only during the equilibration period)

6) Compute averages of the properties sampled.

Pressure

Partition function $Q = \frac{1}{N!\,\Lambda^{3N}} \int_0^L \cdots \int_0^L e^{-\beta U}\, d\vec{r}^{\,N}$ $\qquad L = V^{1/3}$

Helmholtz free energy $A \equiv -k_B T \ln Q$

Pressure $P = -\left(\frac{\partial A}{\partial V}\right)_{N,T}$

$\therefore P = k_B T \left(\frac{\partial \ln Q}{\partial V}\right)_{N,T}$

$= \frac{k_B T}{Q}\left(\frac{\partial Q}{\partial V}\right)_{N,T}$

$$\therefore \left(\frac{\partial Q}{\partial V}\right)_{N,T} = \frac{1}{N!\,\Lambda^{3N}} \frac{\partial}{\partial V}\left\{\int_0^L \cdots \int_0^L e^{-\beta U}\, d\vec{r}^{\,N}\right\}$$

define scaled co-ordinates

$$\vec{s} = \frac{\vec{r}}{L} = \frac{\vec{r}}{V^{1/3}}$$

$$d\vec{s} = ds_x\, ds_y\, ds_z$$

$$= \frac{1}{V}\, dr_x\, dr_y\, dr_z$$

$$= \frac{1}{V}\, d\vec{r}$$

$$\therefore \left(\frac{\partial Q}{\partial V}\right)_{N,T} = \frac{1}{N!\,\Lambda^{3N}} \frac{\partial}{\partial V}\left\{V^N \int_0^1 \cdots \int_0^1 e^{-\beta U}\, d\vec{s}^{\,N}\right\}$$

$$= \frac{1}{N!\,\Lambda^{3N}}\left\{N V^{N-1} \int_0^1 \cdots \int_0^1 e^{-\beta U}\, d\vec{s}^{\,N} + V^N \int_0^1 \cdots \int_0^1 (-\beta)\left(\frac{\partial U}{\partial V}\right)_{N,T} e^{-\beta U}\, d\vec{s}^{\,N}\right\}$$

Substituting in the expression for pressure we get

$$P = \frac{N k_B T}{V} - \frac{\int_0^1 \cdots \int_0^1 \left(\frac{\partial U}{\partial V}\right)_{N,T} e^{-\beta U}\, d\vec{s}^{\,N}}{\int_0^1 \cdots \int_0^1 e^{-\beta U}\, d\vec{s}^{\,N}}$$

ideal contribution

non-ideal contribution (virial)

$$P = \frac{N k_B T}{V} - \left\langle \left(\frac{\partial U}{\partial V}\right)_{N,T} \right\rangle$$

For systems that are pairwise additive

$$U = \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} u(r_{ij}) = \frac{1}{2} \sum_{i=1}^{N} \sum_{\substack{j=1 \\ \neq i}}^{N} u(r_{ij})$$

$$\therefore \frac{\partial U}{\partial V} = \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} \frac{\partial u}{\partial V}$$

$$= \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} \frac{du}{dr_{ij}} \frac{\partial r_{ij}}{\partial V}$$

$$r = \sqrt{x^2 + y^2 + z^2}$$

define $x' = \frac{x}{L}$ $y' = \frac{y}{L}$ $z' = \frac{z}{L}$

$$\therefore r = L\sqrt{x'^2 + y'^2 + z'^2}$$

$$= V^{1/3}\sqrt{x'^2 + y'^2 + z'^2}$$

$$\frac{\partial r}{\partial V} = \frac{1}{3V^{2/3}}\sqrt{x'^2 + y'^2 + z'^2}$$

$$= \frac{V^{1/3}\sqrt{x'^2 + y'^2 + z'^2}}{3V}$$

$$= \frac{r}{3V}$$

$$\therefore \left(\frac{\partial U}{\partial V}\right)_{N,T} = \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} \frac{du}{dr_{ij}} \frac{r_{ij}}{3V}$$

$$= \frac{1}{V} \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} w_{ij} \qquad w_{ij} = \frac{r_{ij}}{3} \frac{du}{dr_{ij}}$$

$$= \text{virial}$$

$$\therefore P = \frac{N k_B T}{V} - \frac{1}{V}\left\langle \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} w_{ij} \right\rangle$$

Lennard-Jones potential

$$u(r) = 4\epsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right]$$

$$\left[\frac{w(r)}{} \quad \frac{}{r} \right]$$

$$\therefore w(r) = \frac{r}{3} \frac{du}{dr}$$

$$= -\frac{4\epsilon}{3} r \left[12 \frac{\sigma^{12}}{r^{13}} - 6 \frac{\sigma^6}{r^7} \right]$$

$$= -8\epsilon \left[2 \left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^6 \right]$$

Chemical Potential

$$\mu = \left(\frac{\partial A}{\partial N}\right)_{V,T}$$

Widom's Test Particle insertion method

$$\mu \simeq A_{N+1} - A_N$$

$$= -k_B T \ln \left(\frac{Q_{N+1}}{Q_N}\right)$$

$$\frac{Q_{N+1}}{Q_N} = \frac{1}{(N+1)\Lambda^3} \frac{\int\cdots\int e^{-\beta U_{N+1}} d\vec{r}^{N+1}}{\int\cdots\int e^{-\beta U_N} d\vec{r}^{N}}$$

$$= \frac{1}{(N+1)\Lambda^3} \int \left\{ \frac{\int\cdots\int e^{-\beta(U_N + \Delta U)} d\vec{r}^{N}}{\int\cdots\int e^{-\beta U_N} d\vec{r}^{N}} \right\} d\vec{r}_{N+1}$$

$$= \frac{1}{(N+1)\Lambda^3} \int_0^V \langle e^{-\beta \Delta U} \rangle \, d\vec{r}_{N+1}$$

$\Delta U \equiv$ change in energy when you insert $N+1^{th}$ particle

important : The averaging is done for a system of N particles. [Probability density $\propto e^{-\beta U_N}$]

In a homogeneous fluid, all positions are equivalent.

Hence the average $\langle e^{-\beta \Delta U} \rangle$ does not depend on $\vec{r}_{N+1}$

$$\therefore \frac{Q_{N+1}}{Q_N} = \frac{\langle e^{-\beta \Delta U} \rangle}{(N+1)\Lambda^3} \int d\vec{r}_{N+1} \rightarrow V$$

$$= \frac{V}{(N+1)\Lambda^3} \langle e^{-\beta \Delta U} \rangle$$

$$\therefore \mu \simeq -k_B T \ln \left(\frac{Q_{N+1}}{Q_N}\right)$$

$$\therefore \beta \Delta \mu = \ln \left(\frac{(N+1)\Lambda^3}{V} \right) - \ln \langle e^{-\beta \Delta U} \rangle$$

Recall from Statistical Thermodynamics

$$\beta \mu = \ln \rho \Lambda^3 + \beta \mu^{(ex)}$$

↑ ideal gas contribution ↑ non-ideal contribution

$$\therefore \beta \mu^{ex} = -\ln \langle e^{-\beta \Delta U} \rangle$$

In Widom's Test Particle technique, the quantity $\langle e^{-\beta \Delta U} \rangle$ is obtained by performing test insertions of the $N+1^{th}$ particle at random locations within the box

Let N_{test} be the number of test insertions attempted per configuration and M be the number of configurations sampled.

$$\therefore \beta \mu^{ex} = -\ln \left[\frac{1}{M} \sum_{m=1}^{M} \left\{ \frac{1}{N_{test}} \sum_{t=1}^{N_{test}} e^{-\beta \Delta U_t^{(m)}} \right\} \right]$$

Typical value of $N_{test} \sim 10^3 - 10^4$

Important : $\Delta U_t \equiv U_{N+1} - U_N$. The insertions are temporary and other

used to compute ΔU_t : they do not affect the Markov Chain

This is an expensive calculation and hence is performed after a certain number of cycles (e.g one once every 10 or 100 cycles)

Pros: One can compute chemical potential in a single simulation

Cons: 1) Only works for moderately dense fluids [at high densities, there is no space to insert a molecule $\Rightarrow e^{-\beta \Delta U_t} = 0$; ΔU_t is large +ve number]
2) Not applicable for crystalline materials